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Practice article

Deep balanced cascade forest: An novel fault diagnosis method for data imbalance

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ABSTRACT

Data imbalance is a common problem in rotating machinery fault diagnosis. Traditional data-driven diagnosis methods, which learn fault features based on balance dataset, would be significantly affected by imbalanced data. In this paper, a novel imbalanced data related fault diagnosis method named deep balanced cascade forest is proposed to solve this problem. Deep balanced cascade forest is a multi-channel cascade forest, in which, each of its channels adaptively generates deep cascade structure and is trained on independent data. To enhance the performance of imbalance classification, the deep balanced cascade forest is promoted from both aspects of resampling and algorithm design. A hybrid sampling method, namely Up–down Sampling, is proposed to provide rebalanced data for each cascade forest channel. Meanwhile, a new type of balanced forest with an improved balanced information entropy for attribute selection is designed as the basic classifier of cascade forest. The good synergy of these two methods is the key to the deep balanced cascade forest model. This good synergy makes deep balanced cascade forest achieve the fusion of data-level methods and algorithm-level methods. Comparative experiments on sufficient imbalanced datasets have been designed to verify the performance of the proposed model, and results confirm that deep balanced cascade forest is much more stable and effective in handling imbalance fault diagnosis problem compared to the popular deep learning methods.

1. Introduction

Rotating machinery plays an utmost important role in the industry [1–3]. Any fault occurring in them would lead to long breakdown time and high maintenance cost [4]. Therefore, the early detection and correct diagnosis of their faults are beneficial to the maintenance and safe application of them [5]. Usually, a rotating machine is healthy in most of its service life. Faults would develop in it but rarely. This is why there is only a small portion of condition monitoring data are collected under the faulty condition of the machine. This results in the so-called data imbalance problem met in the practice of FD [6]. Practices have shown that due to the lack of sufficient fault data samples, the data imbalance brings negative influences on the effectiveness and accuracy of data-driven based FD methods [7]. In IFD, the fault instances are minority classes. How to improve the accuracy of fault diagnosis based on such imbalance data is always an interesting topic worthy to explore.

There are two categories of methods related to the class imbalance problem have been developed. They are data-level methods and algorithm level methods. The typical data-level methods are based on resampling approaches [8], which try to rebalance the number of instances between classes by the means of either over-sampling, under-sampling, or both. In an over-sampling technique (e.g., adaptive synthetic [9] and SMOTE [10, 11]), the number of minority instances is synthesized to narrow the gap between the majority and minority classes. By contrast, an under-sampling technique, e.g., majority under-sampling technique [12], realizes the balance of the classes by removing instead of majority instances. Although these sampling techniques have the potential to improve the situation to a certain extent, they show defects in various aspects. For example, over-sampling encourages overfitting, whereas under-sampling tends to risk the loss of important information [13]. In engineering practice, the imbalance of the data is very serious. The imbalance classification method based on up-sampling, even for generative sampling methods such as SMOTE, are difficult to avoid overfitting problems and as the amount of data and imbalance rate increase, the problem becomes more serious.

As regards the algorithm level methods, many of them have been developed recently by adopting different types of classification models, such as residual learning algorithm [14], kernel fisher discriminant analysis approach [15], optimal margin

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Nomenclature

IFD	Imbalanced data related fault diagnosis
DBCF	Deep balanced cascade forest
FD	Fault diagnosis
SMOTE	Synthetic minority over-sampling technique
ANN	Artificial neural network
gcForest	Multi-grained cascade forest
BF	Balanced forest
Gmean	Geometric mean
AUC	Area under the curve
ROC	Receiver operating characteristic
TP	True positive
FP	False positive
TN	True negative
FN	False negative
CWRU	Case Western Reserve University
NS	Normal state
IRF	Inner race fault
BF	Ball fault
ORF	Out race fault
MFS	Mechanical fault comprehensive simulation experiment system
IR	Imbalance rate
VMD	Variational mode decomposition
IMF	Intrinsic mode function
ELM	Extreme learning machine
SAE	Stacked auto-encoder
SBD	Small balance dataset
IMD	Imbalance dataset
LBD	Large balance dataset
UCI	University of California Irvine
DFCM-MC	Deep Fuzzy C-Mean Clustering
SMLCI	Spectral meta-learner for class imbalance

distribution machine [16], GA-ANN based approach [17], et al. These methods are designed for specific data and problems. The setting of core parameters is the key to the effectiveness of the method, so they have weak generalization ability and heavy manual work. Apart from designing a classification model, Cost-sensitive learnings are also popularly used that deal with the imbalance classification problems with unknown and unidentical costs on algorithmic level [18,19]. They improve the classification accuracy of minorities by increasing the misclassification cost of the minority instances [20]. Although the cost-sensitive methods eliminate the complicated work at the algorithm level, the effect of the method still depends on the setting of the cost weight, which is also a difficulty in complex mechanical systems with unknown data distribution. In general, algorithm level methods are dependent on the problem, which makes it difficult and cumbersome to set reasonable parameters of models [8].

It is found that the application of a single strategy and classification algorithm may lead to unsatisfactory classification results due to their defects. And a machine-learning paradigm called ensemble learning can often perform better than a single method, in which multiple classifiers are trained and used in combination for accomplishing a mission [21]. Based on this idea, this paper integrates data-level methods and algorithm-level methods to improve classification accuracy and stability of imbalance classification. And the key to the application of ensemble learning in the imbalance classification problem is to propose suitable sampling methods and basic classifiers.

Attributing to the simple structure and good performance in classification, various decision tree algorithms show their potential as basic classifiers of ensemble methods [22]. In addition, it has been proved that the decision tree is also an excellent algorithm in building an integrated model for imbalanced data classification [23,24]. And the application of deep learning models in fault pattern recognition has proved to be very effective [25–27]. A novel deep learning method based on decision tree-based ensemble method, i.e., gcForest [28], shows the potential of imbalance classification. So, it is possible to take advantage of gcForest to develop a new deep framework for IFD problems.

However, there are still some issues that need to be addressed in designing a gcForest based IFD model. Firstly, it is difficult for decision trees to deal with high-dimensional data [29]. Random forest will encounter computational efficiency issues when processing high-dimensional condition monitoring signals. Secondly, gcForest does not implement ensemble learning on different data. The gcForest trains all random forests on the same batch of data, i.e. there is only one channel of the data stream in the entire model. However existing research indicates that the positive synergy of sampling and ensemble techniques is beneficial to boost the diversity of training data and improve the accuracy of unbalanced classification [8]. Moreover, the ensemble learning method is reliable as it is based on a substantial piece of theoretical and empirical evidence for supporting the classification [30]. Thirdly, the performance of the basic classifiers on imbalanced data needs to be further improved to assure the accuracy of ensemble learning. The traditional measures for attribute selection of decision trees, such as information gain and gain ratio [31], are highly dependent on data statistics. So the evaluation of attributes is always influenced by data distribution. Usually, the influence of minorities is much smaller than that of the majority.

Based on the above discussion, the DBCF model is proposed in this paper. To reduce the negative influence of imbalanced data on fault diagnosis, an optimized cascade random forest is dedicatedly designed to improve imbalance learning from both data level and algorithm level. The major scientific contributions of this research can be summarized as follows:

(1) A novel multi-channel cascade framework for rotating machine fault diagnosis is proposed, in which a data-level method and an algorithm-level method collaborate to achieve better results.
(2) A hybrid sampling method, namely Up–down Sampling, is proposed to provide rebalanced data for each cascade forest channel. This method reduces over-fitting and improves data diversity by generating new data and discarding samples at the edge of the class distribution.
(3) A new type of balanced forest is proposed as a basic classifier of the cascade forest. The attribute assessment indicator of the balanced forest is a novel balance information entropy, which is calculated based on statistics of the proportion of data.

The remaining part of the paper is organized as follows. In Section 2, the structure of the multi-grained cascade forest is briefly introduced. In Section 3, a Deep Balance Cascade Forest method is proposed. In Section 4, three case studies are given for demonstrating the accuracy of the proposed method in imbalance classification. The influence of imbalanced data distribution on fault diagnosis will be also discussed in this section. The paper is finally concluded in Section 5.

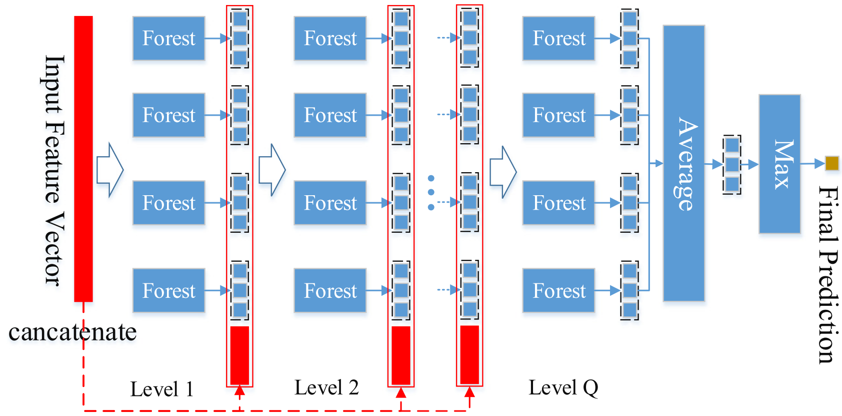

Fig. 1. Illustration of the cascade forest structure.

2. Deep forest

Deep neural networks perform well in representation learning mostly relying on the deep feature produced by layer-by-layer structure. Inspired by this recognition, Zhou and Feng [28] proposed a non-neural network deep model framework that employs a cascade structure. The gcForest consists of two parts, the so-called multi-grained scanning enhances representational learning ability by generating multi-scale features, and cascade structure processes feature information processed layer-by-layer. Similar to the neural network, each layer of the cascade receives the output of the previous layer and output to the next layer after processing. Moreover, the processing of cascade structure is more complicated than the neural network, and an ensemble of random forests is a typical case. However, the most attractive component of gcForest, in this paper, is the cascade forest.

Taking gcForest as an example, each layer of gcForest consists of 4 random forests, the same amount of training data is allocated to each random forest by sampling with replacement. The percentage of different classes of samples at the leaf node is calculated to estimate the class of leaf node, and the class distribution of concerned samples is obtained by counting the tree node where the sample falls into across all trees in the same forest. Class distribution is represented by a probability evaluated class vector with the same dimension as the classes, and then the class distributions of 4 random forests are concatenated with the original data to be the input to the next layer. The last layer averages the probability distribution of all random forests and takes the class with the highest probability as the classification result. Intermediate layers of gcForest are trained as a second-level learner(meta-learner) using the new data generated by the previous layer, this usage of the class vector as a result of the random forest classification is very similar to the idea underlying the stacking method. The architecture of the cascade illustrated in Fig. 1, where each level of cascade receives four 3-dimensional feature vectors concatenated each other and with the original input, and outputs to the next level.

3. The proposed deep balanced cascade forest

In this section, an intelligent fault diagnosis method, namely DBCF, is illuminated in detail. Fig. 2 shows architecture of the proposed method. Fig. 2 shows the data flow of the DBCF method from left to right. The imbalance data is balanced through three steps of Up–down Sampling: calculating the acceptance rate, generating new samples and discarding samples. Up–down Sampling is performed multiple times, and the results correspond to the input of each channel, and each channel is a cascaded balanced forest. The output of multiple channels is synthesized to get the final conclusion. The advantage of the DBCF model lies in the multi-channel deep cascading structure that provides data diversity and an ensemble of classifiers. Pseudo-data is generated by Up–down Sampling to provide a balance dataset, and deviant samples tend to be discarded to avoided overfitting and noise interference. Bagging approach [32] is employed to allocate data to ensemble classifiers, and a multi-channel deep cascade classification model is formed. A new type of BF is proposed as the basic classifier of DBCF. And a novel measure method of attribute selection of decision tree is employed to improve the ability of imbalance classification. The statistics of data are normalized in this novel measure method, and a balance information entropy is calculated based on the proportion of classes to estimate attributes selection. This novel measure method calculates and eliminates the adverse effects of data imbalance in computing information entropy. The cascade model trains and validates each level independently, and keeps growing new level until the accuracy does not improve. The last level BFs' outputs of the deep cascading structure are averaged to obtain the class distribution, and the class with the highest score of the class distribution is regarded as the final result.

3.1. Up–down Sampling

Based on the idea of importance sampling and rejection sampling, an Up–down Sampling method is proposed to alleviate data imbalance. In this new method, the up-sampling and down-sampling will be carried out successively to rebalance the distribution of data and avoid losing the information of minority class.

The reception rate of the sample is defined by the distance between the sample and the other nearest samples. Reception rates, in essence, indicate the degree to which a sample can reflect characteristics of a certain class. Its formula is listed as follows:

$$r_i = \frac{\sum_{j=1}^{n} d(x_i, x_j)}{\sum_{k=1}^{N} d(x_i, x_k)} \tag{1}$$

where r_i is the reception rate of sample x_i, $d(x_i, x_j)$ is Euclidean distance between x_i and x_j, n is defined as the number of samples that belong to the same class as x_i among the N closest samples to x_i.

The up-sampling process generates a new sample between two samples of same minority class, and the distance to the two parent samples is determined by the acceptance rate of the parent sample as follows:

$$x' = \frac{x_i \times r_i + x_j \times r_j}{r_i + r_j} \tag{2}$$

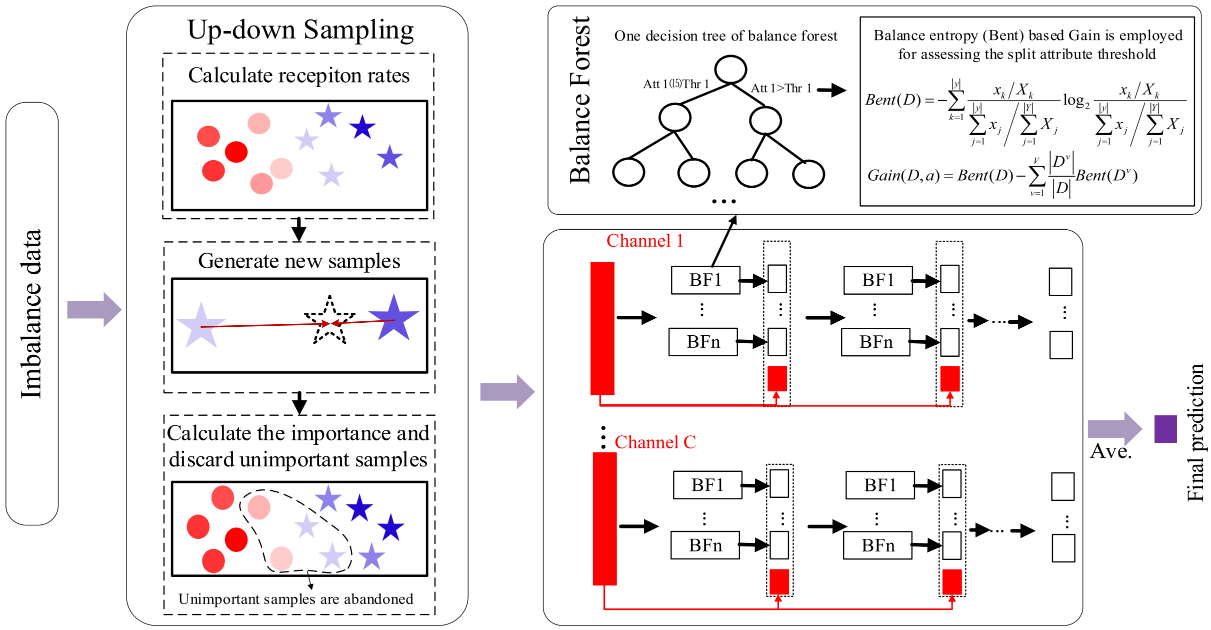

Fig. 2. The structure of DBCF consists of Up–down Sampling and multi-channel deep cascaded balanced forests. Up–down Sampling provides balanced data for the first level of cascade forest, and balance entropy is employed to further optimize the split threshold.

where x' is the generated data, x_i and x_j are randomly chosen samples of minority class, r_i and r_j are accepted rates of x_i and x_j.

Then a series of balance sample sets are generated, and reception rates of all data are recalculated as the sample importance. There is a blurry boundary in the distribution of the generated balance dataset, and down-sampling makes the boundary clear which facilitates efficient classification. Down-sampling is performed by discarding the samples according to probability. The probability of discarding is equal to the reciprocal of importance. The flow of Up–down Sampling is shown in Fig. 3.

The process of Up–down Sampling is performed according to probability, hence the difference and diversity of the results of each Up–down Sampling are guaranteed. In the DBCF method, Up–down Sampling was executed C times to provide datasets for C channels. This scheme makes it possible to apply the Bagging idea to the deep cascading deep forest structure and improve the diversity of the classifier and data, which is significantly helpful for dealing with the imbalance problems.

Actually, Up–down Sampling does not guarantee that the distribution of output data is completely balanced. The down-sampling process aims to retain the samples that are closer to the center of the class distribution, nonetheless, Up–down Sampling effectively alleviates data imbalance, and the diversity of minority class data is certainly improved. Therefore, the basic classifiers are not fed with absolute balance data as training data.

3.2. New type of balanced forest

The basic classifier of DBCF, proposed in this paper, is a new type of balanced forest. Satisfactory performance of classifiers is critical for guaranteeing the positive role of the bagging strategy. However, the imbalanced data distribution disturbs the attribute selection of decision trees, which will deteriorate the performance of trees. As a widely used criterion for expressing the degree of confusion of data distribution based on statistics of instances, information entropy is obviously going to be affected by imbalance data.

To ensure that the attribute selection measure of decision trees is not affected by the imbalanced data, an improved information entropy, namely balance information entropy, is employed in the new type of balanced forests, i.e.

$$Bent(D) = -\sum_{k=1}^{|y|} \frac{\frac{x_k}{X_k}}{\frac{\sum_{j=1}^{|y|} x_j}{\sum_{j=1}^{|Y|} X_j}} \log_2 \frac{\frac{x_k}{X_k}}{\frac{\sum_{j=1}^{|y|} x_j}{\sum_{j=1}^{|Y|} X_j}} \tag{3}$$

where $|y|$ is the number of classes of current branch, $|Y|$ the number of total classes, x_k the number of samples which belong to class k in set D, X_k the number of samples belong to class k. The trees of BF adopts Info-Gain Ratio split criterion, where the Info-Gain Ratio equals to:

$$Gain(D, a) = Bent(D) - \sum_{v=1}^{V} \frac{|D^v|}{|D|} Bent(D^v) \tag{4}$$

where a is a split threshold of attribute, V is the number of split branches, $|D^v|$ is the number of samples of branch v. In this paper, $V = 2$.

3.3. Procedure of DBCF

The DBCF method comprises two modules, i.e., Up–down Sampling and the multi-channel cascade forest. Up–down Sampling is for generating balance training data for each channel of the multi-channel cascade forest module. The training set of each channel will be separated into two parts: growing set and estimating set. The cascade grows during the training process on the growing set until the performance on the estimating set does not improve. Then the number of levels of each channel is obtained adaptively which is not necessarily equal in each channel. The results of the channels are averaged to get the final outputs. The more the number of channels of multi-channel cascade forest, the higher the diversity of the classifier is, and the better the ability to solve complex classification problems tends to be. However, a more complex model is less efficient and prone to develop overfitting issues, so the number of channels should be determined based on the complexity of the classification problem. In this paper, DBCF is suggested to be designed with two channels by trial and error. The implementing steps in detail are described in Alg. 1.

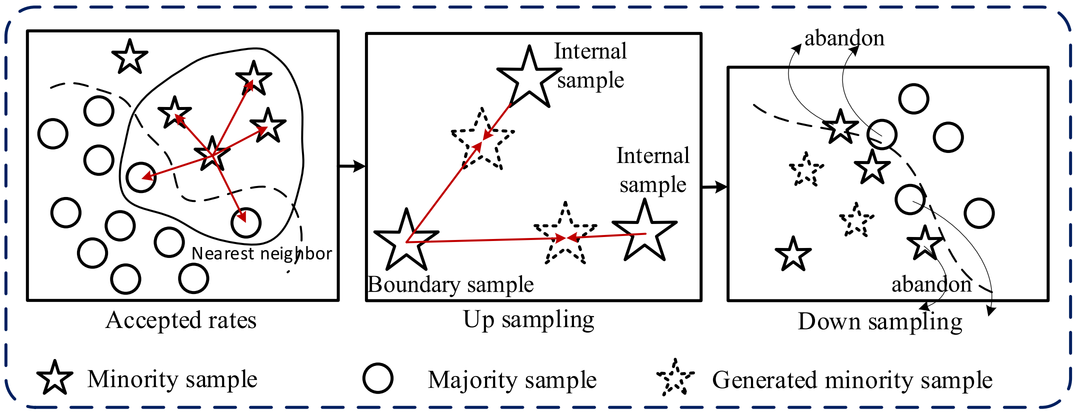

Fig. 3. The Up sampling process generates new data and the Down sampling process discards data.

Algorithm 1 The procedure of DBCF

Initialize parameters: the number of nearest neighbors in the up-sampling S, the number of channels C, the length of the input L, the number of balanced forests in each cascade structure N, and the maximum depth of the cascade Q.

Training phase

Input: Train set S_{train}.

for the channel i **do**
  Execute Up-down Sampling to obtain train set $S^{(i)}$ of the channel i according to (1) and (2).
  Separate $S^{(i)}$ into a growing set $S^{(i)}_{growing}$ and estimating set $S^{(i)}_{estimating}$ equally.
  Set the current depth $q_i = 1$.
  while not converge and $q_i < Q$ **do**
    Train the balance forests with $S^{(i)}_{growing}$ according to (3) and (4).
    Test the balance forests with $S^{(i)}_{estimating}$.
    Output the result of $S^{(i)}_{growing}$ and $S^{(i)}_{estimating}$ to next layer.
    $q_i \leftarrow q_i + 1$
  end while
end for
Save the DBCF model.

Testing phase

Input: Test set S_{test}.

for the channel i **do**
  Test the DBCF with S_{test}.
  Average the output of last layer as the output of channel i.
end for
Average the outputs of all channels to get a class distribution vector V.
Take the maximum value of V as the final conclusion.
Output: The label corresponding to the maximum value of V.

4. Experimental verification

4.1. Evaluation metric

To assess the imbalance classification result, two metrics, i.e., Gmean and AUC, are employed in this research.

Gmean is the geometric mean of all class accuracies and is more preferable to evaluate the effectiveness of classifiers for class imbalance learning [33]. Gmean is defined as below:

$$Gmean = \sqrt{\frac{TP}{TP+FN} \times \frac{TN}{TN+FP}} \tag{5}$$

Gmean is the geometric average of positive and negative accuracy, which illustrates the accuracy from two aspects.

ROC curve as a performance measure for machine learning algorithms is a good way of visualizing a classifier's performance. The curve describes the relationship between the true positive rate and false positive rate under different classification thresholds. AUC is equivalent to the probability of correctly distinguishing the positive and negative samples and the larger the AUC the better the performance of the classifier [33]. AUC can be obtained

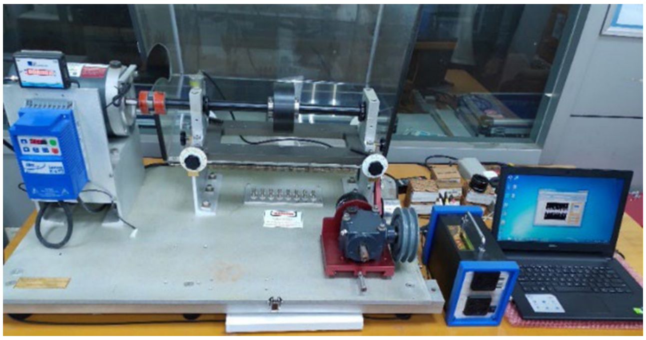

Fig. 4. Rolling bearing fault test rig.

by

$$AUC = \frac{\left(1 + \frac{TP}{TP+FN} - \frac{FP}{FP+TN}\right)}{2} \tag{6}$$

4.2. Experimental data description

In this section, two kinds of datasets will be used for verifying the proposed method. The Electrical Engineering Laboratory of CWRU provided a common dataset, which contains four kinds of status, including NS, IRF, BF, and ORF. The motor speeds are 1797 to 1720 RPM with different motor loads.

Another dataset was collected by MFS in our laboratory, shown in Fig. 4, MFS is a piece of experimental equipment manufactured by SpectraQuest Company in the United States to simulate the bearing failure of rotating machinery.

In this study, the experimental dataset consists of 9 CWRU datasets with different imbalance rate, load and failure degree, and 15 MFS datasets with different imbalance rate, loads and rotation frequencies, as listed in Table 1. D1–D24 are the serial numbers of dataset1–dataset24. The majority class refers to the normal state, minority class refers to each fault class. IR refers to the imbalance rate of dataset, i.e. the ratio of data sizes of majority class and minority class.

4.3. Preprocessing: Feature extraction

As mentioned earlier, DBCF will suffer efficiency and training challenges when fed with high-dimensional features. In this paper, VMD [34] is employed to decompose signals into IMFs. This method is less sensitive to noise and sampling than the aforementioned methods [35]. Then three types of features are combined: time domain features, frequency domain features and modal features based on IMFs. Mode feature includes energy entropy feature, singular value features and corresponding center frequency of each IMF. Specifically, 23 time and frequency domain features are selected as suggested in [36]. The energy entropy features are obtained as suggested in [37]. The singular value is the natural characteristic of the matrix and possesses beneficial stability [36]. Singular value features will be used to reduce the dimension and express decorrelation characteristics as suggested in [38].

As presented in Fig. 5, the feature vector consists of 12 time-domain features, 11 frequency-domain features, and 2K+1 VMD modal features. Each feature vector is calculated from raw vibration data with a length of 1200, and data from CWRU and MFS was sampled at 12K. Each input contains 0.1 s of vibration data. Bearing rotation frequency is ranging from 10 Hz to 30 Hz, corresponds to data of 1 to 3 rotation cycles respectively.

4.4. Case I: Comparison with fault diagnosis methods

In comparison, the ANN is trained by BP learning algorithm [39], called "ANN". ELM method is formed with the wavelet active function [40], called "WELM". Additionally, the oversampling approach SMOTE [10] is combined with stacked auto-encoder SAE [23], called "SMOTE-SAE".

The parameters K and alpha of VMD are 5 and 2500 respectively. According to the calculation method described in Section 4.3, each input feature vector will contain 34 features. The parameters of the methods are detailed in Table 2.

Table 3 shows the result of four methods on three evaluation indexes, and the maximum value of each evaluation index in the four methods are highlighted using bold font. Dataset D20–D24 are the datasets with the largest amount of data and the largest number of classes. Dataset D14, D19 and D24 are the datasets with the highest degree of imbalance. All the evaluation indexes of DBCF on these complex datasets are the best compared with the other three methods. The accuracy, AUC and Gmean verified on datasets with IR over 10 were analyzed, and 83.33%, 91.67% and 91.67% of optimum values are results of DBCF. The average is shown in the last row, the proposed method is much better than the other three methods which means that the proposed method is more accurate and reliable in dealing with the imbalance classification problems.

To visualize the performance of different levels of complexity problems, all the experimental results are illustrated in a three-dimensional scatter plot, as shown in Fig. 6. It is visible in Fig. 6(a) that the sample points of DBCF are gathered closely and are located near the maximum value of the coordinate axis. In Fig. 6(b), (c) and (d), the distribution of the results of the four methods in terms of accuracy, AUC and Gmean are similar, which shows that the comparison of the four methods is stable from various angles. The red line in the box plot represents the average evaluation of the model, and the vertical width of the box plot indicates the stability of the model on different datasets. It is seen that the red lines of DBCF are higher and boxes of DBCF are much narrower than other methods. This fully proves the excellent classification capability of DBCF.

4.5. Case II: Comparison with decision tree based methods

Regarding the data imbalance problem, this study aims to establish an accurate and reliable method for diagnosing bearing faults. An anomaly was found in the experiments, existing methods perform better with SBD, even if the overall number of the dataset is less than the imbalance dataset. In this part, new datasets were selected to verify this problem, which was listed in Table 4. The amount of data for each class of SBD is the same as the amount of data for minority classes of IMD, and the amount of data for each class of LBD is the same as the amount of data for majority classes of IMD. The volume of data and IR designed inconsistent to compare the impact of volume and distribution on classification. There are 2 balance datasets corresponding to each imbalanced data, and the data volume of the 2 balance sets are equal to the majority and minority class of the imbalance set respectively.

Random forest and gcForest are both decision tree based classifiers. Herein, they will be compared with DBCF for proving the advantage of DBCF on imbalance classification. To ensure a reliable comparison, the maximum of layers of DBCF and gcForest is set to be 5, and the number of trees is 100 whereas the tree number of random forest is 500.

Table 5 summarizes the experimental results in case II. Comparing the results of SBD1, IMD1 and LBD1, DBCF performed noticeably well on imbalance data, and similar to the other two

Table 1
Data used in the comparison between fault diagnosis methods.

No.	Majority	Minority	IR	Data description
D1	60	30	2	4 classes drive end vibration data. Consists of NS, IRF, centered ORF and BF with 0 load and fault diameter is 0.007 inches.
D2	60	10	6	
D3	60	5	12	
D4	60	30	2	12 classes drive end vibration data. Consists of NS, IRF, centered ORF and BF with 0 load, fault diameter is 0.007, 0.014, 0.021, 0.028 inches and centered ORF lacks the data of 0.028 inches fault.
D5	60	10	6	
D6	60	5	12	
D7	60	30	2	13 classes dataset. Consists of IRF, ORF, BF and NS, loads range from 0 to 3, fault diameter is 0.007 inches. The load of Normal state data is 0.
D8	60	10	6	
D9	60	5	12	
D10	500	100	5	5 classes vibration data. Consists of NS, IRF, ORF, BF and CF with 0 load, and the rotational frequency of the axle is 10 Hz.
D11	600	50	12	
D12	900	100	9	
D13	900	50	18	
D14	900	10	90	
D15	500	100	5	9 classes vibration data. Consists of NS, IRF, ORF, BF and CF, fault states contain 0 and 1 load, and rotational frequency of the axle is 10 Hz. The load of Normal state data is 0.
D16	600	50	12	
D17	900	100	9	
D18	900	50	18	
D19	900	10	90	
D20	500	100	5	13 classes vibration data. Consists of NS, IRF, ORF, BF and CF with 0 load, fault states contain 3 rotational frequencies ranging from 10 Hz to 30 Hz. The speed of NS data is 10 Hz.
D21	600	50	12	
D22	900	100	9	
D23	900	50	18	
D24	900	10	90	

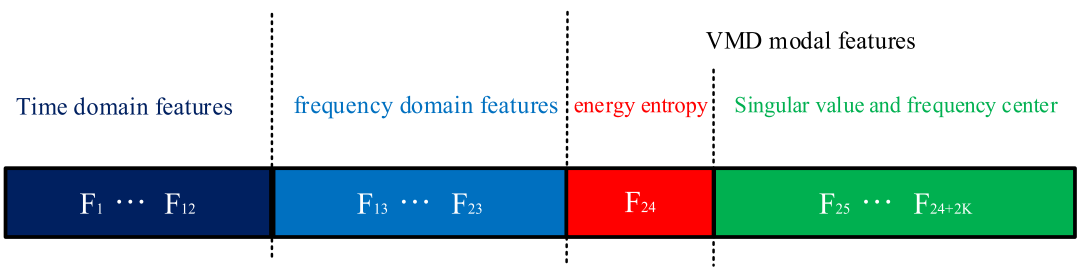

Fig. 5. The Multi-scale Feature consists of 12time domain features, 11 frequency domain features and 2K+1 VMD modal features.

Table 2
Parameters used in case 1.

Method	Width of layers	Depth of layers
ANN	34-512-256-128-64-(class number)	6
WELM	34-512-(class number)	3
SMOTE-SAE	34-512-256-128-64-(class number)	6
DBCF	4 BFs per layer, BF consists of 200 decision trees.	Adaptively grow. Maximum is 5.

methods on balance data. Details are shown more obviously in Fig. 7. According to the bad performance of gcForest and RF on imbalanced data, it can be verified that the distribution of data is also an important factor of classification. Neglecting this issue may lead to worse performance after increasing data of certain classes, as the balance of the dataset may be disrupted.

To more scrupulously compare the impact of IR and data volume on model results, Fig. 8 shows a comprehensive comparison of results. Fig. 8 contains a comparison of two quantities, namely data volume and IR. Along the axis "data volume" and axis "IR" direction can independently compare the influence of the data volume and IR on the model performance. It can be found that both the data volume and the IR have an impact on the model results, while gcforest and RF are more affected by IR. Comparing the three histograms in each row, we can find that gcForest and RF both have the worst performance on imbalance data even if the data volume of imbalance dataset is the same as the balance dataset. However, the experimental results of DBCF are close for different IRs, which proves that DBCF effectively improves the performance and stability of dealing with data imbalance problems. It can be concluded that the damage of the imbalanced data distribution to the classification is ubiquitous, and the current method cannot cope with this problem, the proposed DBCF method takes this issue into consideration and performs well.

4.6. Case III: Comparison with other imbalance classification methods

Considering that DBCF is an ensemble deep learning method involving sampling, deep feature learning and neural network theory, some frontier synthetic oversampling and machine learning methods proposed recently were selected for comparison in this part. WMODA [7] and IMH [41] employed data resampling methods to solve imbalance problem, and WMODA generates synthetic samples. CSKELM [33] combines a cost-sensitive method with extreme learning machines to simplify calculations. DNCNN [24] is a typical deep learning method which adopts weighted softmax loss to handle imbalanced classification problem. In addition, two semi-supervised and unsupervised methods, i.e., DFCM-MC [42] and spectral meta-learner for class imbalance

Table 3
Results in case 1.

No.	IR	ACC (%)				AUC				Gmean			
		DBCF	ANN	WELM	SMOTE-SAE	DBCF	ANN	WELM	SMOTE-SAE	DBCF	ANN	WELM	SMOTE-SAE
D1	2	**100.00**	**100.00**	**100.00**	**100.00**	**1.000**	**1.000**	**1.000**	1.000	**1.000**	**1.000**	**1.000**	1.000
D2	6	**100.00**	**100.00**	**100.00**	100.00	**1.000**	**1.000**	**1.000**	1.000	**1.000**	**1.000**	**1.000**	**1.000**
D3	12	**100.00**	75.00	**100.00**	75.00	**1.000**	0.914	**1.000**	0.667	**1.000**	0.704	**1.000**	0.704
D4	2	**99.80**	90.60	94.20	85.60	**1.000**	0.956	0.993	0.910	**0.999**	0.907	0.966	0.877
D5	6	**98.50**	90.60	91.90	69.80	**0.999**	0.943	0.991	0.819	**0.992**	0.907	0.952	0.708
D6	12	**99.00**	62.90	92.30	66.50	**0.999**	0.872	0.985	0.784	**0.994**	0.628	0.954	0.688
D7	2	**92.30**	91.30	51.70	64.80	**0.997**	0.932	0.799	0.967	**0.953**	0.915	0.663	0.755
D8	6	76.90	**82.70**	38.70	57.70	**0.988**	0.897	0.767	0.936	**0.830**	0.829	0.582	0.692
D9	12	**77.70**	66.90	53.30	54.20	**0.987**	0.831	0.806	0.843	**0.835**	0.694	0.672	0.650
D10	5	92.50	**93.00**	24.50	86.00	**0.995**	0.989	0.671	0.965	0.952	**0.953**	0.321	0.910
D11	12	**88.50**	84.50	51.00	85.00	**0.988**	0.973	0.810	0.954	**0.927**	0.895	0.648	0.902
D12	9	90.00	**94.50**	29.50	87.50	**0.991**	0.989	0.717	0.966	0.937	**0.964**	0.413	0.920
D13	18	**89.50**	85.50	26.50	86.00	**0.990**	0.927	0.716	0.939	**0.934**	0.894	0.269	0.909
D14	90	**81.50**	41.50	65.50	68.50	**0.970**	0.702	0.854	0.883	**0.881**	0.395	0.754	0.779
D15	5	80.00	**81.10**	51.10	69.20	**0.980**	0.983	0.819	0.920	**0.879**	0.837	0.683	0.812
D16	12	**71.10**	57.80	39.20	61.40	**0.967**	0.934	0.710	0.913	**0.816**	0.633	0.585	0.752
D17	9	75.30	**82.80**	45.00	61.10	0.974	**0.987**	0.802	0.848	0.843	**0.846**	0.621	0.714
D18	18	**75.60**	63.30	38.10	61.40	**0.972**	0.954	0.731	0.917	**0.851**	0.678	0.571	0.757
D19	90	**61.40**	11.10	37.20	45.60	**0.950**	0.402	0.730	0.834	**0.746**	0.000	0.553	0.634
D20	5	**94.60**	85.40	74.40	90.60	**0.999**	0.993	0.932	0.988	**0.969**	0.881	0.841	0.946
D21	12	**91.70**	88.70	64.60	84.40	**0.998**	0.970	0.892	0.983	**0.953**	0.899	0.782	0.909
D22	9	**92.90**	85.80	74.80	85.20	**0.998**	0.940	0.937	0.986	**0.959**	0.883	0.849	0.916
D23	18	**92.10**	7.70	66.20	88.50	**0.997**	0.428	0.906	0.988	**0.955**	0.886	0.793	0.934
D24	90	**81.90**	69.80	64.80	68.30	**0.993**	0.928	0.885	0.866	**0.892**	0.719	0.785	0.741
mean		**87.62**	74.69	61.44	75.10	**0.989**	0.894	0.852	0.943	**0.921**	0.789	0.719	0.817

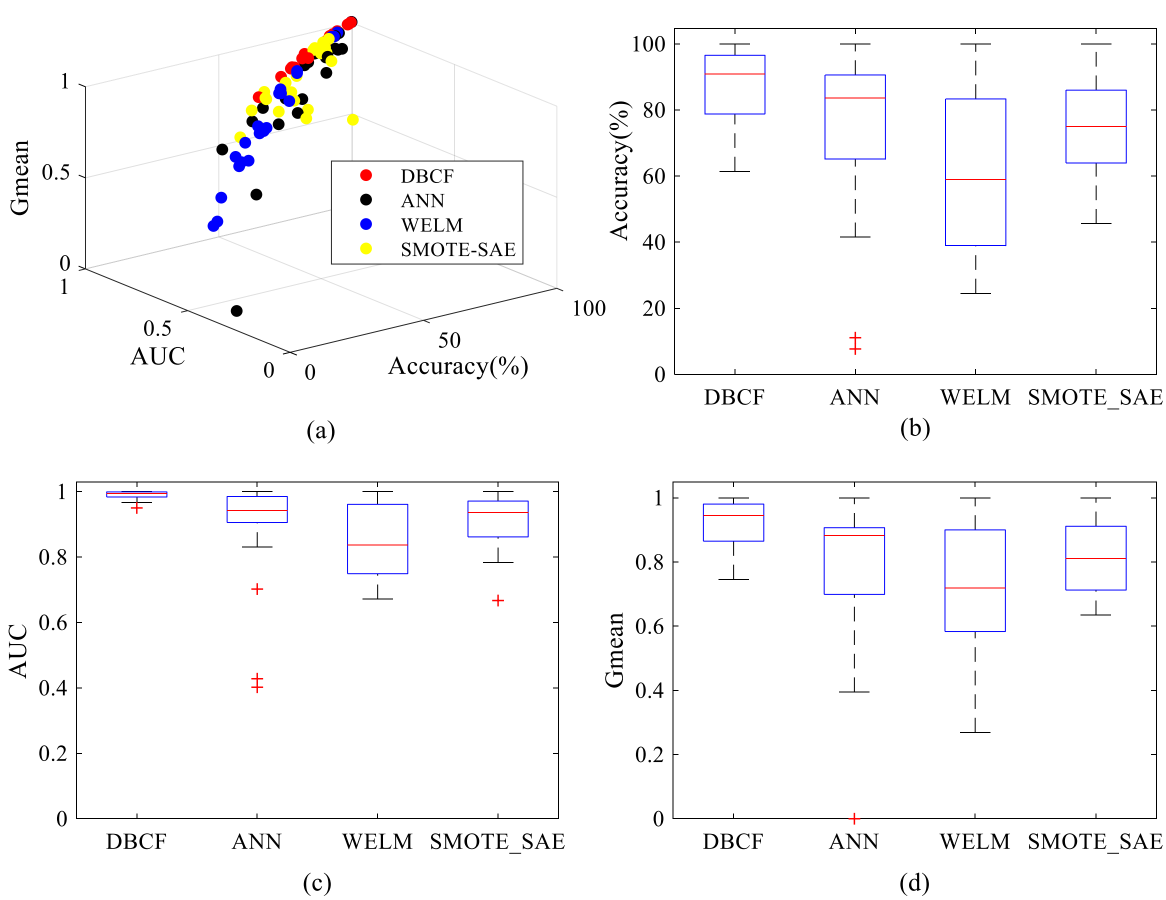

Fig. 6. Results of the 24 dataset on different methods. (a) is the 3-dimensional scatter plot of three indexes and reflects the distribution of overall experimental results. (b), (c) and (d) are the box plots of accuracy, AUC and Gmean, which reflect the stability of each method.

(SMLCI) [43], were taken into consideration. DFCM-MC addresses the multi-class imbalanced issue by deeply decomposing semi-supervised data into multi-clusters. SMLCI is an unsupervised heterogeneous ensemble learning approach, which uses cost-sensitive learning. Recently, some scholars combined the generative confrontation network and the convolutional neural network

Table 4
Date used in case 2.

No.	Class	Majority	Minority	IR	Data description
LBD1	12	20	20	1	CWRU datasets. Consists of N, IRF, centered ORF and BF with 0 load and the speed is 10 Hz, fault diameter is 0.007, 0.014, 0.021, 0.028 inches and centered ORF lacks the data of 0.028 inches fault.
IMD1	12	20	5	6	
SBD1	12	5	5	1	
LBD2	13	200	200	1	MFS datasets. Consists of N, IRF, ORF, BF and CF with 0 load, fault states contains 3 speeds ranging from 10 Hz to 30 Hz. The speed of Normal state is 10 Hz.
IMD2	13	1400	100	10	
IMD3	13	280	20	10	
SBD2	13	40	40	1	

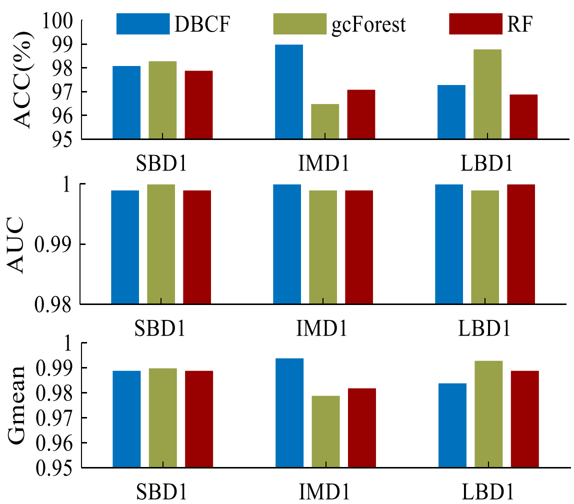

Fig. 7. Compare the effects of the three methods on SBD1, IMD1 and LBD1, DBCF shows obvious advantages on IMD1, and gcForest and RF performs worst on IMD1 than SBD1.

Table 5
Results of case 2.

No.	ACC			AUC			Gmean		
	DCBF	gcforest	RF	DCBF	gcforest	RF	DCBF	gcforest	RF
LBD1	0.981	**0.983**	0.979	0.999	**1.000**	0.999	0.989	**0.99**	0.989
IMD1	**0.990**	0.965	0.971	**1.000**	0.999	0.999	**0.994**	0.979	0.982
SBD1	0.973	**0.988**	0.969	**1.000**	0.999	**1.000**	0.984	**0.993**	0.989
LBD2	**0.956**	0.952	0.939	0.996	**0.998**	**0.998**	**0.975**	0.973	0.971
IMD2	**0.941**	0.905	0.892	**0.996**	0.986	0.985	**0.969**	0.957	0.955
IMD3	**0.897**	0.89	0.861	**0.995**	0.984	0.981	0.932	**0.934**	0.919
SBD2	**0.938**	0.935	0.913	**0.998**	0.997	0.993	0.966	0.963	**0.973**

Table 6
UCI data used to test DBCF in case 3.

Dataset	No. of Samples	No. of Classes	Max IR	Min IR
Wine	59/71/48	3	1.47	1.2
Page-Blocks	2000/329/28/88/115	5	71.42	6.08

to propose an enhanced network (GAN-CNN) to solve the imbalanced classification problem and validated on the CWRU bearing dataset [44].

For a fair comparison, three datasets similar to those in the above literature were added to validate the effect of DBCF in CASE III, the details are shown in Table 6. The datasets of WMODA and DNCNN are obtained on fault bearing test rigs that are similar to the MFS; the datasets of CSKELM, DFCM-MC and SMLCI are selected from the UCI machine learning repository dataset; and the datasets of IMH and GAN-CNN are selected CWRU dataset. Several datasets with approximate IR were selected from the literature of these methods for comparison. All the corresponding evaluation

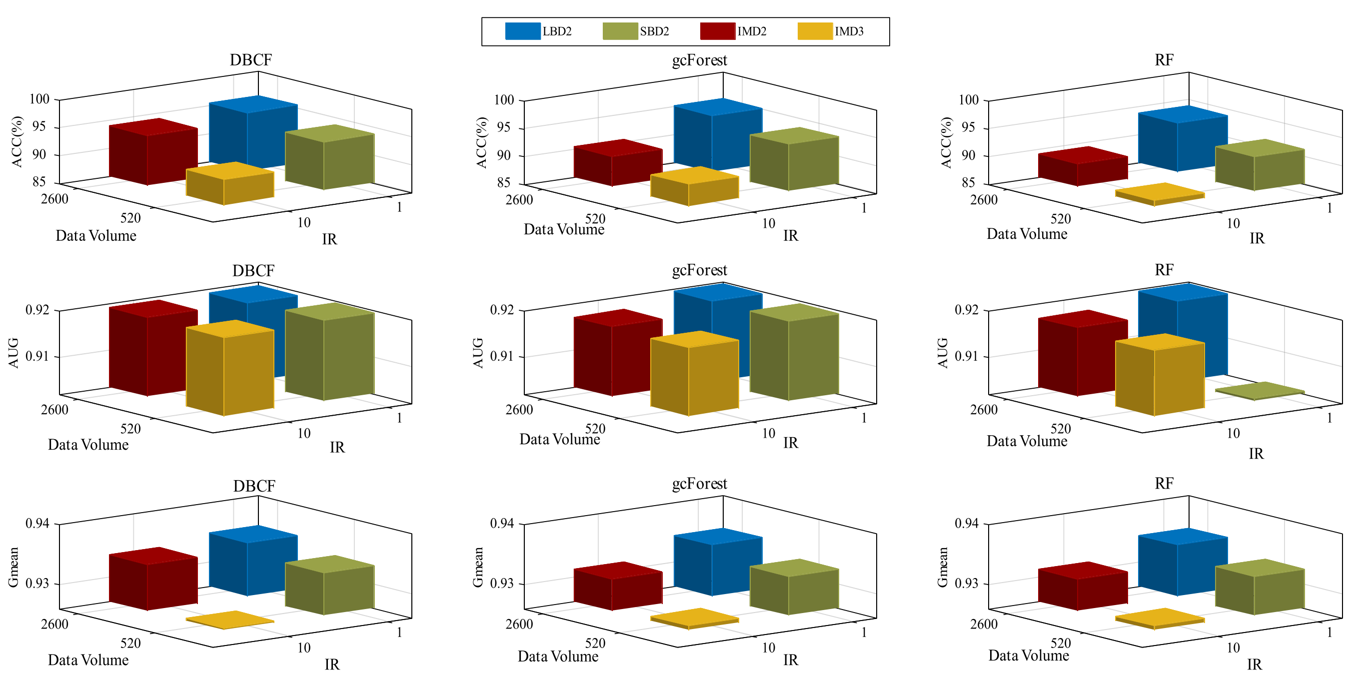

Fig. 8. Compare the result on dataset LBD2, IMD2, IMD3 and SBD2, which correspond to two different IR and two different data volume respectively. Each column in Fig. 8 corresponds to one certain method, and each row corresponds to one certain evaluation index. As can be seen, the increase of IR and the decrease of data amount have a bad influence on the result, but DBCF is less sensitive to IR than the other two methods.

Table 7
Comparison between different advanced methods.

Method	Dataset	Class number	IR.	Accuracy (%)	AUC	Gmean
WMODA	fault bearing datasets	2	1.47	\	\	0.91
			5.36	\	\	0.91
			7.25	\	\	0.853
DNCNN		8	1	99.22	\	\
			1.67–5	98.19	\	\
			2.5–25	95.52	\	\
IMH	CWRU datasets	4	**1.25**	**98.58**	\	**0.986**
			5	**97.12**	\	**0.971**
			10	**97.91**	\	**0.979**
GAN-CNN		10	**2**	**100**	\	\
			5	**99.31**	\	\
			10	**95.58**	\	\
DFCM-MC	UCI datasets	3	**1.3**	**0.987**	**0.993**	\
		10	11.65	60.39	0.8866	\
		8	15.27	78.02	0.9531	\
		5	31.64	93.26	0.9762	\
SMLCI		2	1.3	98.66	\	\
			10.74	84.88	\	\
			11.11	88.69	\	\
CSKELM		2	2.53	\	1	1
			8.77	\	**0.962**	**0.939**
			39.22	\	0.788	0.754
DBCF	CWRU datasets	4	**6**	**100**	**1**	**1**
			12	**99**	**0.999**	**0.994**
		12	**2**	**99.8**	**1**	**0.999**
			6	**98.5**	**0.999**	**0.992**
			12	**99**	**0.999**	**0.994**
	MFS datasets	13	18	92.1	0.997	0.955
			90	81.9	0.993	0.892
	UCI datasets	3	**1.2-14.7**	**100**	**1**	**1**
		5	**6.08-71.42**	**96.6**	**0.986**	**0.908**

indicators of the proposed method are listed in Table 7. To ease comparison all data are sorted by the IR size. Obviously, DBCF is superior to these existing methods in the case of more classes and higher IR. And experimental performance on these three types of datasets illustrates the application potential of DBCF in various fields. Specifically, comparing the experimental results of IMH and DBCF on CWRU dateset (black bold font), in the case that the number of categories is the same and the degree of imbalance in the data processed by DBCF is higher, the effect of DBCF is better than IMH; comparing the results of WMODA, CSKELM and DBCF on the dataset "Page-Blocks" from UCI (red bold font), DBCF keeps the results close to other two methods with more classes and higher imbalance rate; comparing the results of DFCM-MC and DBCF on the dataset "Wine" from UCI (blue bold font), DBCF has better performance with higher imbalances. The results of DBCF on D4, D5 and D6 can be compared with the results of GAN-CNN, they come from same data source, and the data size of GAN-CNN is 3.33 times of DBCF, and the class number and IR of GAN-CNN is less than DBCF. DBCF and GAN-CNN perform similarly when IR is less than 10 and DBCF shows obvious advantages when IR is more than 10, indicating that DBCF surpasses existing methods in solving the problem of high imbalance.

4.7. Discussion on model efficiency

Model efficiency of the proposed DBCF is analyzed in this subsection, and comparison methods in Case I are involved for comparison. The experiment in Case I uses a PC with Intel Core i7-9700 CPU (8 cores), the experimental results about efficiency are listed in Table 8. SMOTE-DAE contains the up-sampling process, the training data size is much larger than the other three methods. Obviously, its efficiency is low, and this is the natural disadvantage of up-sampling based methods, so it is not included in the comparison in this experiment. Besides, the efficiency of ANN method is affected by two training parameters, that is, "epoch" and "batch size". Generally, the time required for training is positively correlated with epoch and inversely correlated with batch size. After trial and error, the optimal parameters of ANN without considering efficiency issues are that epochs are around 200, and the batch sizes are between 10 and 50. Thus the results of ANN with epoch equals 200, and the batch sizes equal 10 and 50 are listed in Table 8.

According to Table 8, DBCF does not have advantages on training efficiency, the training time of DBCF is larger than ANN with batch size equaling to 50 and less than ANN with batch size equaling to 10. However, compared with ANN which needs to adjust parameters for different data, DBCF has reliable and stable performance, which will save a lot of labor costs in practical applications. The efficiency of WELM is the best, which is a simple three-layer neural network structure method, however it is very poor in dealing with complex problems. In addition, the test time cost of DBCF is about 0.0035 s per sample (The average result of combining all the data) while that of ANN is 0.0003 s per sample (The average result of combining all the data).

5. Conclusion

An innovative DBCF model is proposed as an application to rolling element bearings fault diagnosis in this study. The main motivation behind DBCF is the need to design an efficient method for imbalance fault diagnosis. DBCF combines the idea of deep cascade forest and optimizes the sampling method and the threshold evaluation function of the decision tree. The proposed model was experimentally verified, positive synergy works between Up–down Sampling which generates, and screens diversity data based on the distribution of data and random forest based on balance information entropy. The average AUC and Gmean reached 0.989 and 0.921, which is improved by an average of 9.07 and 13.57 compared with control methods. In this manner, the faults could be more accurately and steadily evaluated under imbalanced data conditions. In addition, the effect of imbalanced data on classification is discussed, experimental results

Table 8
Comparison of efficiency between different methods.

No.	Majority	Minority	IR	DBCF	ANN		WELM
					batch=10	batch=50	
D1	60	30	2	10.01	34.59	5.37	4.15
D2	60	10	6	5.41	19.37	4.47	2.50
D3	60	5	12	4.28	14.12	3.81	1.47
D4	60	30	2	61.30	113.45	17.80	12.03
D5	60	10	6	24.36	42.03	5.61	4.09
D6	60	5	12	14.69	24.59	4.29	3.14
D7	60	30	2	68.61	125.17	21.41	15.82
D8	60	10	6	28.65	44.30	5.86	13.86
D9	60	5	12	16.61	27.51	4.91	6.41
D10	500	100	5	133.56	285.84	73.92	11.93
D11	600	50	12	112.29	263.29	62.00	9.84
D12	900	100	9	192.94	570.96	136.51	21.44
D13	900	50	18	154.60	465.39	100.25	16.29
D14	900	10	90	116.83	350.04	93.85	12.68
D15	500	100	5	239.32	495.05	108.68	21.72
D16	600	50	12	151.53	323.12	75.29	14.25
D17	900	100	9	276.20	782.88	175.19	34.25
D18	900	50	18	181.36	495.41	110.36	22.00
D19	900	10	90	94.37	294.39	67.75	13.39
D20	500	100	5	287.77	551.40	110.26	34.70
D21	600	50	12	185.25	359.09	73.34	19.57
D22	900	100	9	334.27	823.14	184.36	53.85
D23	900	50	18	223.18	473.12	112.73	28.74
D24	900	10	90	124.92	234.62	67.27	14.86

have shown that the classification is not only dependent on the quantity of data, but is also affected by the uneven distribution of the data. The existing studies have not fully considered these problems, but they have been successfully solved by the proposed DBCF method. In summary, the proposed DBCF model applied a deep cascade frame to rotating machinery fault diagnosis, which could effectively deep feature mining and stable classification of imbalanced fault data.

Despite the achievement described above, more effort is still required to further improve the proposed method in the future. For example, in the present form of this method, BF consists of two-branch decision trees, which can be replaced with multi-branch decision trees. And an adaptive method is also needed to determine the number of branches. In addition, compared with the existing neural network model, the training efficiency of DBCF has no advantage. Since the training method of decision tree is less efficient than matrix operation. On the other hand, DBCF lacks a suitable parallel computing framework, which also makes it more efficient in practical applications. Far below the neural network.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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